

ISAAC JIANG

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EDUCATION & ACADEMIC PERFORMANCE

Cornell University

Ithaca, NY

Class of 2028,

Bachelor of Science, Mechanical Engineering

Courses: Statics and Mechanics of Solids, Thermodynamics, Differential Equations for Engineers, Multivariable Calculus for Engineers, Physics I: Mechanics & Heat, Physics II: Electromagnetism, Intro to Experimental Physics

RELEVANT EXPERIENCE

Engineering Intern || Queens College Internship,

June 2025 - August 2025

- Assisted research and design of an automatic recirculating sump system for a new fish tank facility, supporting 21 fish tanks.
- Designed and assembled a return and drainage PVC piping system for the circulation of water into the tank's sump system.
- Utilized Fusion 360 to model a sensor float that housed sensors to monitor the water conditions within the recirculating system.

Liquid Propulsion @ Cornell || Mechanical Subteam Member,

April 2025 - Present

- Modeled a composite vessel in SolidWorks to withstand 800 PSI, applying FEA simulation to validate structural integrity.
- Contributing to comprehensive research, feasibility study, and initial design phase for a 10kN amateur-class turbopump.
- Developing the preliminary CAD architecture for the turbopump, currently focusing on initial designs of the gas generator.

AguaClara Project Team || Ram Pump Subteam Member,

September 2025 - Present

- Investigating the operational principles and fluid dynamics of a sustainable gravity-powered vertical ram pump system.
- Contributing to snifter valve integration to enhance pump efficiency and longevity by re-aerating the system air chamber.
- Assisted in designing various piping configurations to optimize system layout, improve hydraulic efficiency, and simplify assembly for field implementation.

Co-founder & Mentor || Unity Scholars Network,

October 2024 - Present

- Co-founded and created a network providing free, personalized mentorship to underrepresented high school students in NYC.
- Currently actively assisting mentees amidst the application season, equipping them with the necessary resources and support.

Biology Research Intern || Queens College, College Now STEM research

July 2023 - August 2023

- Contributed to three written and published scientific experiments; experienced wet and dry lab experimentation and fieldwork.
- Involved in monitoring and surveying various fish tanks; experienced in scoring experimentation recordings for numerical data.
- Assisted in the repair, test, and calibration of experimental equipment to gain visual data of experimental planktonic crustaceans.

PROJECTS

Modular sensor float

June 2025 - August 2025

- Assisted in the brainstorming, design, and modeling of a modular sensor float that monitored the health of the sump system.
- Drafted and prepared technical deliverables like project designs and models, along with progression reports and calculations to ensure initial design works as intended.
- Implemented sensor float capable of housing 10 interchangeable sensors in any orientation, maximizing experimental flexibility.

Modeled COPV(Composite Overwrapped Pressure Vessel)

April 2025 - May 2025

- Modeled a fiberglass COPV in Solidworks to withstand high internal pressures of 800 PSI on the internal faces.
- Tested the model of the COPV in FEA simulation using engineering techniques such as static analysis and stress distributions.

EXTRACURRICULAR ACTIVITIES

- Member, Guild of Visual Arts (2024-Present)
- Member, Cornell Taekwondo (2024-Present)
- Artist/Painter, hobbyist (2020-Present)
- Member, Chinese Student Association (2024-Present)
- Member, Society of Asian Scientists and Engineers (2024-Present)

SKILLS AND CERTIFICATION

Technical: Autodesk Fusion 360 Certified User, Fusion 360, Solidworks, Onshape, Python, JavaScript

Soft: Adaptability, Attention to detail, Critical thinking, Communication, Teamwork